

RESEARCH

Open Access



Improving the performance of PV/diesel microgrids via integration of a battery energy storage system: the case of Bilgo village in Burkina Faso

Daniel Yamegueu^{1*} , Henri Thomas Nelson² and Amidou Singho Boly³

Abstract

Background PV/diesel microgrids are getting more popular in rural areas of sub-Saharan Africa, where the national grid is often unavailable. Most of the time, for economic purposes, these hybrid PV/diesel power plants in rural areas do not include any storage system. This is the case in the Bilgo village in Burkina Faso, where a PV/diesel microgrid without any battery storage system has been set up. This power plant is composed of three diesel generators operating in parallel (two of 16 kW and one of 24 kW), coupled with a photovoltaic field of 30 kWp. It was observed that for such power plants, the grid management is not always efficient due to constantly fluctuating solar output and loads. This inconsistency in energy output raises the question if integrating battery energy storage systems could improve the grid's performance. While many studies in the literature focus on hybrid energy systems, only a few of them have tackled the optimization of existing and operational systems.

Methods This study investigated three scenarios based on the existing microgrid's characteristics: conventional standalone diesel generators, PV/diesel without battery storage and PV/diesel with a battery storage system which are the main technologies used for off-grid rural electrification in Burkina Faso. The levelized cost of electricity (LCOE) was used to assess the economic performance of each scenario, and the calculations were made using the HOMER software.

Results It was found that the best among the scenarios considered is the PV/diesel/battery configuration which has the lowest LCOE of US\$ 0.524/kWh. The battery storage system for the optimal configuration has a capacity of 182 kWh with about 8 h of autonomy.

Conclusions It can be inferred from this study that a storage unit is necessary for an optimal management of a PV/diesel microgrid. Indeed, the storage unit significantly reduces the operating and maintenance costs associated with running diesel generators, as well as the excess electricity. The storage system also allows for a greater reduction in CO₂ emissions compared to systems without storage.

Keywords Microgrid, Diesel generator, Photovoltaic system, Energy management, Battery storage, Levelized cost of electricity

*Correspondence:
Daniel Yamegueu
daniel.yamegueu@2ie-edu.org; dan.yamegueu@gmail.com
Full list of author information is available at the end of the article



© The Author(s) 2024. **Open Access** This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>. The Creative Commons Public Domain Dedication waiver (<http://creativecommons.org/publicdomain/zero/1.0/>) applies to the data made available in this article, unless otherwise stated in a credit line to the data.

Background

Access to reliable electricity is essential for the socio-economic development of any country. In sub-Saharan Africa, the electricity access rate is very low, which negatively impacts the region's economic growth and living standards [1].

According to the International Energy Agency (IEA), 757 million people remain without electricity worldwide in 2020, 584 million of them are living in Africa [2]. In Burkina Faso, as of 2020, the national electricity access rate stood at 19%, with 69% in urban areas and less than 1% in rural areas [2]. Those with access to electricity in most African countries face numerous challenges, such as unreliable power supply and high electricity costs. However, this situation is more critical, particularly in rural areas, where projected scenarios indicate that over 700 million people will still lack access to electricity in 2040 [3].

Renewable energy sources have been identified as the most suitable alternative to fossil fuel sources for power generation in most developing countries. They are abundant in nature and environmentally friendly compared to fossil fuel sources [4, 5]. They can help reduce fuel costs and challenges due to technical and economic constraints associated with grid expansion systems [6].

Nevertheless, generating electricity solely from single renewable sources, such as wind or solar photovoltaic, comes with several drawbacks. These include high initial investment costs and low power reliability due to the intermittent and uncertain nature of these sources, particularly when relying on a single energy source [7]. These challenges can be overcome by adopting hybrid renewable energy systems (HRES). HRES combines two or more electricity-generating sources, such as renewables and conventional energy sources, to produce reliable energy. In this concept, the different energy sources complement each other when generating power from the system to supply loads [8].

PV/diesel hybrid systems without battery storage units, based on the *flexy energy* concept, have been developed and implemented for electricity generation in off-grid areas, especially in Burkina Faso and Mali [9, 10]. As shown in previous studies cited below, battery storage was excluded in the *flexy energy* concept to reduce the replacement cost in the system and the environmental concerns associated with batteries at the end of their lifetime.

Azoumah et al. [9] performed a simulation of three power-generating scenarios, namely diesel generators only, a PV/battery system, and a PV/diesel without battery system. From their analysis, they reported that the PV/diesel without battery system scenario was the most optimal system among the three considered in terms

of LCOE and CO₂ emissions. They concluded that PV/diesel hybrid systems based on the *flexy energy* concept could be a better alternative in rural and peri-urban areas if their design management is improved.

Yamegueu et al. [11] carried out experimental work on a PV/diesel system without a battery storage system. The study assessed the behavior of the PV/diesel hybrid system for different ranges of load profiles, representing different nominal power levels of the diesel generator, namely 40%, 62%, 82%, and 105%. It was found that the contribution of the PV array affects the output performance of the diesel generator because the generator was operating at a lower-rated capacity, which results in high fuel consumption and consequently to a high electricity production cost.

Tsuanyo et al. [12] carried a techno-economic modeling on PV/diesel hybrid system without battery storage. In their study, two cases were considered. The first one is composed of a PV system coupled with identical diesel generator capacities, while the second case comprised a PV system coupled with different diesel generator capacities. Both cases were compared with conventional diesel generators. The study presented two potential optimization solutions. The first solution involved using seven identical diesel generators with a capacity of 26 kW each, which were paired with a PV generator of 120 kWp. In the second solution, three diesel generators with capacities of 35 kW, 54 kW, and 75 kW were coupled with a PV generator of 150 kWp. The LCOE for the two optimal solutions was found to be 0.285 €/kWh when compared with to the conventional diesel generators, which was found to be 0.32 €/kWh.

However, eliminating batteries from the PV/diesel system has its drawbacks, such as the excess electricity produced by the system is not put into any productive use, stability problems, extended operating hours, high operation and maintenance (O&M) costs, and a short lifetime for diesel generators.

Many similar studies have been carried out over the recent years on hybrid power generation systems for decentralized applications. Most of these studies mainly focused on assessing the techno-economic feasibilities and viability, performance optimization, and reliability of power plants. In prominent studies [8, 13–19], the authors have determined the best hybrid configuration for off-grid electrification, in the respective contexts of Benin, Chad, Ethiopia, Iraq, Malaysia, Nigeria, Thailand, and Zambia, based especially on the net present cost (NPC) and LCOE parameters. However, to the best of our knowledge, very few studies are focused on the optimization of a real existing and operational system. The present study investigates the integration of a battery energy storage system (BESS) to an existing hybrid

off-grid hybrid energy system to optimize its operation. Although BESS could have great advantages in microgrid systems, they also have some disadvantages. Specifically, size and cost are highly important since a high capacity causes increases in cost and size, while low capacity may not be enough to prevent unexpected power problems and meet load demand. Consequently, BESS size must then be carefully calculated to determine the optimum size for a given system.

This study contributes to understanding the feasibility and viability of implementing PV/diesel microgrids with integrated battery energy storage by conducting a comprehensive analysis based on economic and environmental parameters. The village Bilgo in Burkina Faso has been considered as case study. The village has been chosen because it already hosts a PV/diesel microgrid without storage built in the framework of the ACP-EU Energy Facility's "flexy-energy" project (Reference 129–364). It is therefore a good case study for highlighting the advantages of integrating storage into a microgrid that does not contain any.

The microgrid is composed of three diesel generators operating in parallel (two of 16 kW and one of 24 kW), coupled with a photovoltaic field of 30 kWp.

The paper is arranged into five sections. Following this introduction, the study methodology is presented, followed by results and discussion sections. The conclusion wraps up the paper.

Methods

Case study area

Bilgo village is the selected location for this study. It is located in the Pabre commune, a section of Kadiogo

Province in the Central Region of Burkina Faso. The village is about 30 km from the city capital Ouagadougou and lies between latitude $12^{\circ} 31.8$ N and longitude $1^{\circ} 40.8$ W. According to the 5th general population and household census report [20], there were about 2483 inhabitants in Bilgo in 2019. The primary sources of household earnings in Bilgo village are agricultural activities and livestock rearing.

Access to electricity in Bilgo village was a challenge before the installation of the PV/diesel power plant in 2016 due to the absence of an extension of the electricity grid reaching the village. Most households relied on inefficient lighting sources such as touch lights, oil lamps, candles, and small thermal generators. The problem of electricity access in the village was a major setback towards the quality of living, better health services, quality education, water supply, and small businesses. The commissioning of the power plant is expected to stimulate development in the village.

The village has an abundant solar resource (Fig. 1), thus it is well suited for the installation of solar PV systems.

The solar resource data were obtained from the National Aeronautics and Space Administration (NASA) database for the given latitude of Bilgo village. The monthly solar radiation for the site location ranges between 5.09 to 6.43 kWh/m²/day, with an annual average solar radiation of 5.76 kWh/m²/day. The average clearness index for the location was found to be 0.59, which indicates a good potential for PV system applications.

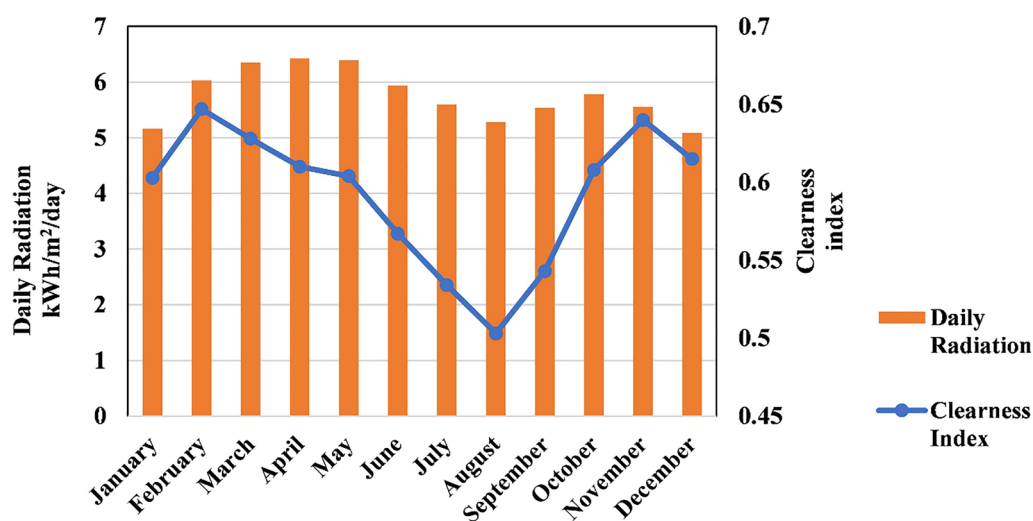


Fig. 1 Solar resource in Bilgo village

Study power plant

This study is based on the existing PV/diesel without battery storage system installed in Bilgo village. Based on the previous feasibility study, this hybrid power plant was designed to meet the electrical load demand in the village. The power plant relies on the contribution from both power sources for the load profile of the village. It comprised a 30 kW_p solar PV capacity, subdivided into five arrays (three with 5 kW_p and two with 7.5 kW_p). The PV arrays are coupled with three diesel generators with a total capacity of 56 kW, two of 16 kW (DG1 and DG2) and one of 24 kW (DG3). The parallel operation of this hybrid system enables either of the power-generating sources to supply as per the demand. Moreover, it allows the system to operate simultaneously during peak demand.

The configuration of the PV/diesel hybrid setup is illustrated in Fig. 2

The five PV arrays are all connected to DC protection buses to ensure the safety interruption of electricity flowing from the PV array to the inverters. The DC buses are connected to five grid-tied inverters (three with 5 kW capacity and two with 7 kW capacity). The inverters are also connected to multiple circuit breakers to ensure maximum system protection. The measuring meters are used to measure the voltage, current, and power output from the inverter before they are connected to the AC bus. The three diesel generators

are directly connected to the AC bus since its output is AC. The AC bus is connected to a central control unit, which contains multiple components that control the entire operation of the power plant.

Scenarios considered for the study

A comparison between the existing power plant configuration and two other configurations has been conducted. Then, three scenarios have been considered to meet the same load profile in Bilgo village, which are: the diesel generator (DG), the PV/diesel and the PV/diesel/battery systems. These scenarios are widely used for off-grid rural electrification in Africa, especially in Burkina Faso [11]. The schematic diagram of the different scenarios considered for the study is shown in Fig. 3.

The characteristics of the systems considered in the present study are:

- The existing power plant: PV/diesel generators (56 kW, comprising two 16 kW and one 24 kW diesel generators, and a PV array of 30 kW_p) without battery storage.
- Three conventional diesel generators (two of 16 kW and one of 24 kW).
- PV/diesel/battery (56 kW, 30 kW_p) with a sensitivity analysis on the storage capacity.

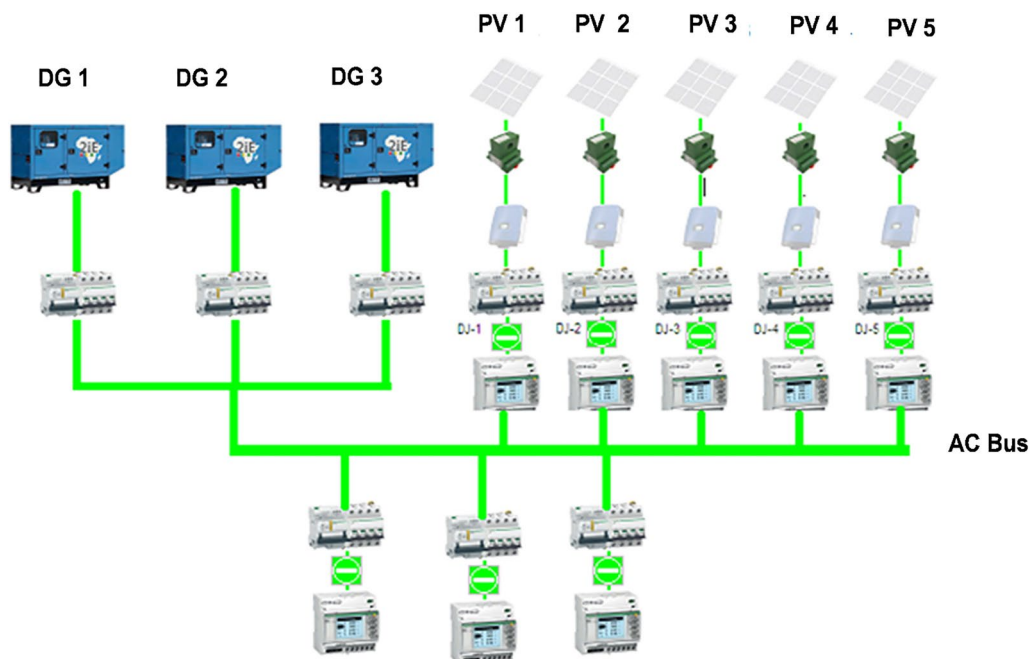


Fig. 2 Architecture of the PV/diesel hybrid configuration of the Bilgo power plant

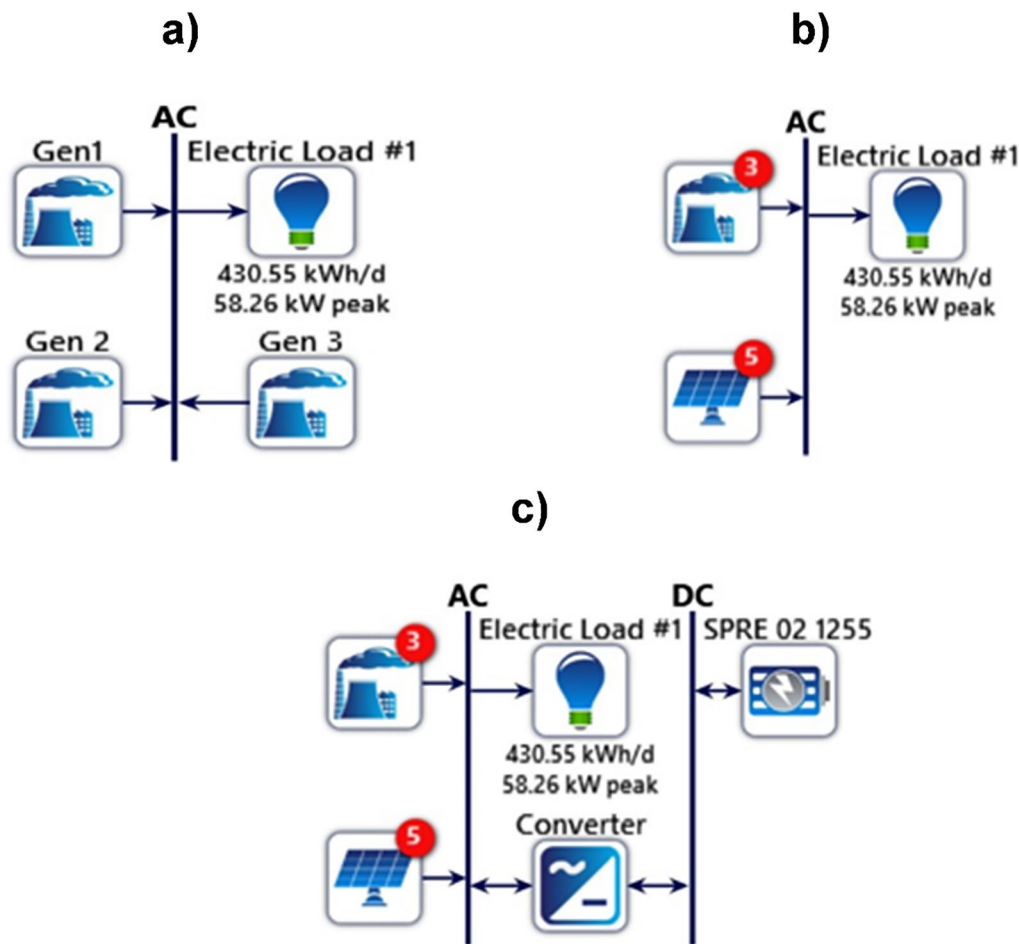


Fig. 3 Schematic representation of the different scenarios considered: **a** diesel generator only, **b** PV/diesel, **c** PV/diesel/battery

Optimization methodology

The HOMER Pro 3.10 software (created by the National Renewable Energy Laboratory, USA) was used for simulations. HOMER uses a time resolution of 1 h for system operation. This resolution allows HOMER to examine daily variability while being able to quickly run many scenarios. The optimal configuration is determined by considering various factors such as the NPC, the LCOE, the capacity shortage, the excess electricity production, the CO₂ emission, and the renewable energy fraction (RF). The LCOE parameter is one of the most relevant and frequently used to compare different energy technologies concerning their costs. The LCOE method can reflect the key factors of the production cost throughout the lifetime of the power plant in just one number and it causes a great reduction in complexity and allows a quick and easy comparison of different alternatives.

Load assessment

The daily load profile was obtained from an evaluation of the electrical needs of the village (see Fig. 4). The daily energy consumption is estimated at 430.55 kWh/d with a peak load of 47.636 kW from 20:00 to 21:00 in the evening. The load demand is high between 18:00 to 00:00 because almost all the villagers are at home, and most of their electrical appliances are operating. The low load demand periods occur during the day between 14:00 and 16:00, when most of the villagers are not at home and most appliances are not in use.

For the simulation process in this study, a day-to-day and time-step variability of 5% was taken into account. This was done to introduce randomness to the load profile and make it more representative of real data across the entire year. An operating reserve of 10% on the diesel generator nominal power has been considered to satisfy the random variability of the load and solar production.

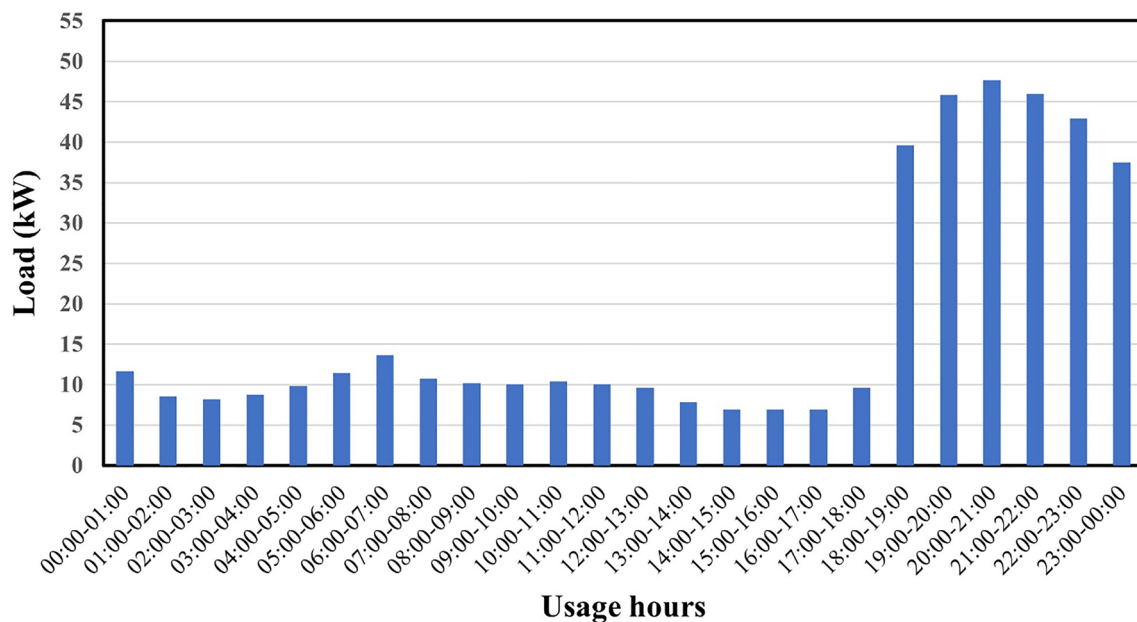


Fig. 4 Daily load profile in Bilgo village

This provides a maximum safety margin to ensure the reliability of the electricity supply regardless of the sudden change in electrical loads or sudden decrease in renewable energy output [21].

Components assessment and costing

The components considered in this study are solar PV modules, converters, diesel generators, controllers, and batteries. The technical data for all the components were based on the manufacturer's specification, and the costs of components such as PV modules, diesel generators, grid-tied inverters, and controllers were obtained from the cost they were acquired during the commissioning of the project.

PV module assessment

The PV array consists of 120 polycrystalline flat plates with a rating of $250 W_p$ each. The total rating of the PV modules is $30 kW_p$. The modules were inclined at a slope of 15° relative to the location's latitude. The azimuth angle of the PV module was zero, and the panels were oriented toward the south to capture the maximum amount of solar radiation during the day. The lifetime of the modules is 25 years, and the derating factor is 85%, which accounts for all the losses due to the effect of temperature, wiring losses, shading, and ageing, which reduces the power output of the panels when compared to the rated power of the PV panel. The capital and replacement costs of the modules were estimated to be US\$1062.61/

kW, including shipping cost, installation, and taxes. The O&M cost was considered 1% of the total capital cost.

Inverters and DC protection assessment

The grid-tied inverters were used to convert the DC output from the PV arrays to the AC bus. In the existing system, each grid-tied inverter was connected to an array to ensure a full power supply to the AC bus. The use of grid-tied inverters for this system aims to reduce the number of combiner boxes and enhance easy troubleshooting and system monitoring [22]. The selected inverters used in this study are SMA TriPower 5000 TL and 7000 TL [23, 24]. The inverters' rated input voltage (V_{mpp}) is 580 V, with a maximum voltage (V_{max}) of 1000 V. The maximum efficiency of the inverters is 98%. The capital and replacement costs of the inverter, including DC protection, were estimated at US\$1770.76 /kW, and the O&M cost of the inverter is assumed to be US\$ 10/year, with an inverter operational lifetime of 15 years.

Diesel generator assessment

Diesel generators in hybrid power systems are usually sized to meet the peak load demand [25] when other renewable sources fail to meet the demand. The minimum load ratio of the generator was set at 20%, and the operating lifetime of the diesel generators was 15,000 h. The capital and replacement costs for the diesel generators are US\$1894.55/kW, and the O&M cost is US\$ 0.02/h.

System control unit assessment

The control system is composed of different components (transducers, circuit breakers, regulators, and generator controllers). It controls, regulates, and creates a means of communication between all the components involved in the hybrid system to ensure the smooth operation of the power plant. The dispatch strategy used in this study is the *load following strategy*. The total capital and replacement costs of the system control unit, including all other embedded components, amount to US\$ 59,292.75. Furthermore, the annual O&M cost is US\$ 100. The lifetime of the system controller is 25 years.

Battery storage unit assessment

The Trojan SPRE 02 1255 was the selected battery technology for the simulation. Each battery has a nominal voltage of 2 V, a nominal capacity of 1270 Ah, and an efficiency of 80%. The capital (including shipping cost and installation cost) and replacement cost of one battery were US\$ 600, and the O&M cost was considered as 1% of the capital cost.

System converter assessment

The selected converter used in the simulation is an SMA Sunny Island 8.0 H bi-directional inverter with a total rated capacity of 30 kW and an efficiency of 95.8%. The converter has a DC input voltage of 48 V, and the maximum charging current is 140 A. The capital cost (including installation cost) and replacement cost of the system converter is US\$ 900/kW, and the O&M cost was 1% of the capital cost. The lifetime of the converter is 15 years.

Economic input parameters

The lifetime of the project has been taken as 25 years. The discount rates in Burkina Faso were estimated to be between 6 and 10% according to previous research [12, 26]. The escalated inflation rates range between 1 and 3% [27]. For the study, the discount rate and the inflation rate will be taken as 8% and 3%, respectively, and the real discount rate is estimated at 4.85%. The price for diesel oil in Burkina Faso was about US\$ 0.99/L (CFA 577.88 per liter) [28]. Since the country is a net importer of petroleum products from neighboring countries, it is more vulnerable to oil price fluctuation in the world market. During scarcity in the country, the fuel price in rural areas is drastically increasing, making it more expensive to operate a power plant running only with diesel engines.

System simulation constraints

For a smooth operation of the system, specific conditions must be satisfied when modeling a hybrid power system. The software rejects the systems during optimization if

these specific conditions are not met. For this study, two main constraints were considered, namely the operating reserve and the maximum annual capacity shortage. The operating reserve is the surplus operating capacity that instantly responds to a sudden increase in the electrical load demand or a decrease in the output power from renewable energy sources. For this study, an operating reserve of 10% was considered as recommended by Adaramola et al. [25] to ensure a reliable power supply for the given load profile. The capacity shortage is the shortfall between the actual operating capacity and the actual amount of operating capacity the system can provide, while the maximum annual capacity shortage is a total shortfall that occurs throughout the year. The accepted maximum allowable annual capacity shortage reported by Adaramola et al. [25] ranges from 0.5 to 5%. In this study, to ensure that there is no shortfall, the maximum capacity shortage of 1% is considered.

Results

Conventional diesel generators only (Scenario A)

In scenario A (see Fig. 3a), three conventional diesel generators (two of 16 kW and one of 24 kW) are used to supply power to the given load profile in Bilgo village. The results obtained show that the annual electricity production by the power plant is equal to 157,145 kWh/year, with a contribution from each generator represented in Fig. 5. The diesel generators 1 and 2 (DG1 and DG 2) have the nominal power of 16 kW each and the diesel generator 3 (DG 3) has a nominal power of 24 kW.

The total annual capacity shortage for this scenario is equal to 194 kWh/year due to the shortfall between the required operating capacity and the capacity supplied by the system. The LCOE in this scenario is US\$ 0.5816/kWh, with a total NPC of US\$ 1,307,126.00 (see Fig. 6). The system's O&M cost over the project's entire lifetime is US\$ 65, 942.43, with a capital investment of US\$ 165,387.80. As the power plant solely operates on diesel engines, the annual fuel consumption for electricity production is 51,232 L. This fuel consumption represents approximately 55.5% of the total system cost. The specific fuel consumption for the three diesel generators (DG1, DG2, and DG3) are 0.271 L/kWh, 0.343 L/kWh, and 0.447 L/kWh, respectively. The lifetime for the three diesel generators is found to be 6.65 years for DG1, 1.73 years for DG2, and 7.28 years for DG3. The annual CO₂ emissions from the system are estimated at 134,116 kg/year due to the long operating hours of the diesel generators.

PV/diesel generator configuration (Scenario B)

Scenario B is composed of a PV/diesel hybrid system without battery storage (see Fig. 3b). This configuration

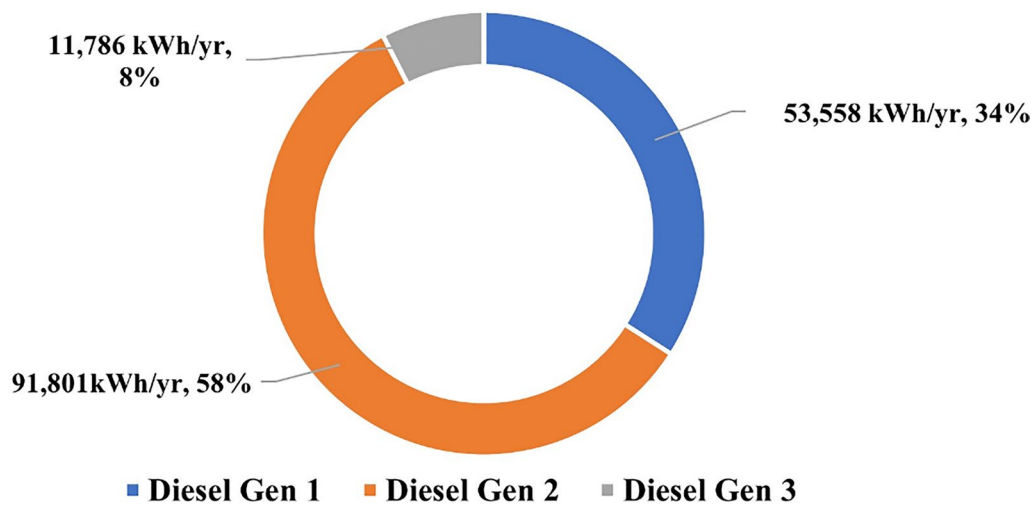


Fig. 5 Total electricity production by diesel generators annually

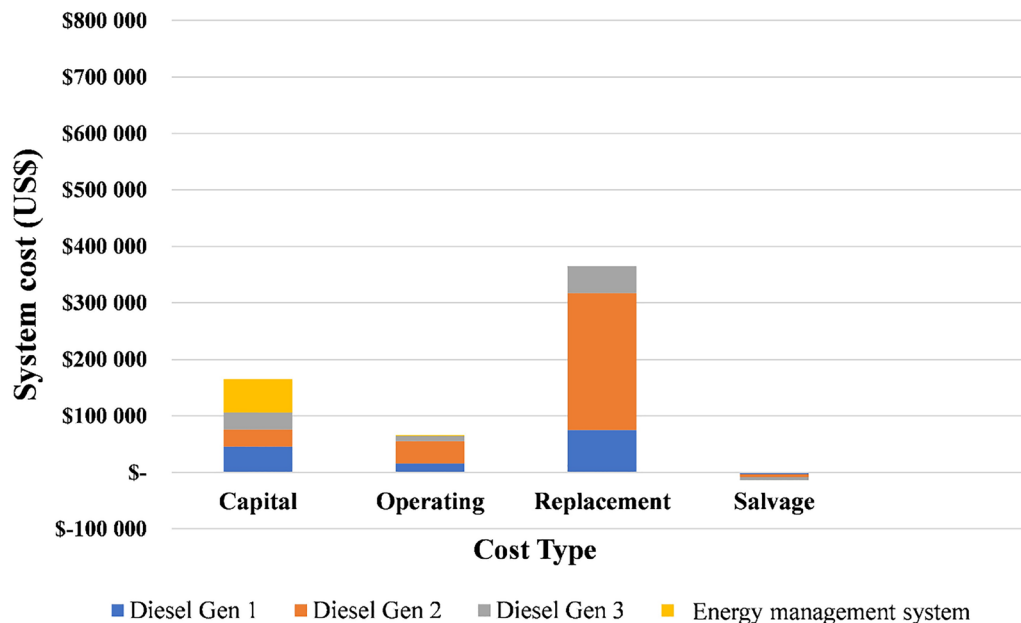


Fig. 6 Total NPC for scenario A

illustrates how the renewable energy source influences the electricity production from the power plant. The results show that the total annual electricity production amounted to 186,877 kWh/year. The electricity production by the PV array and the diesel generators are 48,406 kWh/year and 138,472 kWh/year, respectively (see Fig. 7), with a renewable fraction of 25.9%.

The annual excess electricity production by the system is 29,732 kWh/year, which exceeds the load demand of the village by 15.9%. The annual capacity shortage is 194 kWh/year.

The LCOE for this scenario is US\$ 0.604/kWh, with a total NPC and operating cost of US\$ 1,357,160 and US\$ 74,341.70, respectively. The estimated annual fuel consumption of diesel generators is 46,932 L per year, representing approximately 49% of the total system cost. The specific fuel consumption by the three generators (DG1, DG2 and DG3) are 0.271 L/kWh, 0.333 L/kWh, and 0.49 L/kWh, respectively. The amount of CO₂ emitted by the system is estimated at 122,871 kg/year. This emission is primarily attributed to the diesel generators, as the PV arrays do not release any pollutants into the atmosphere.

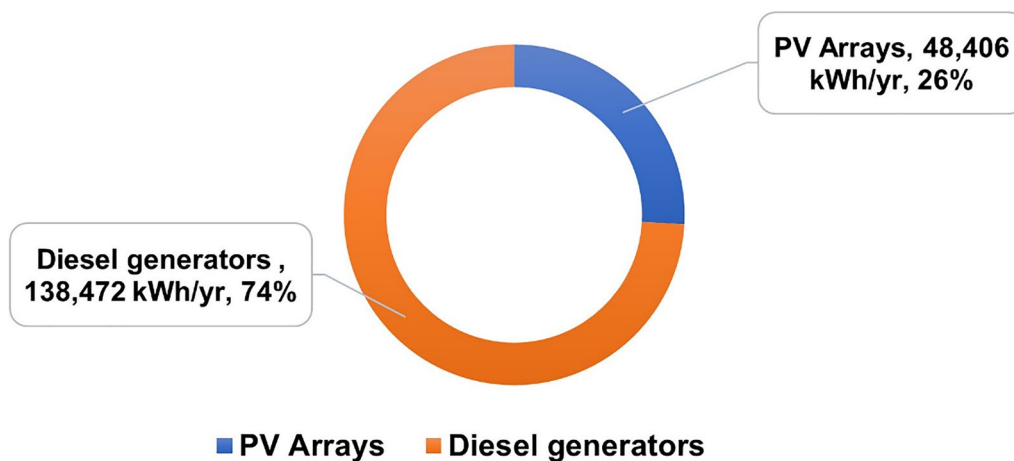


Fig. 7 Total annual electricity production for scenario B

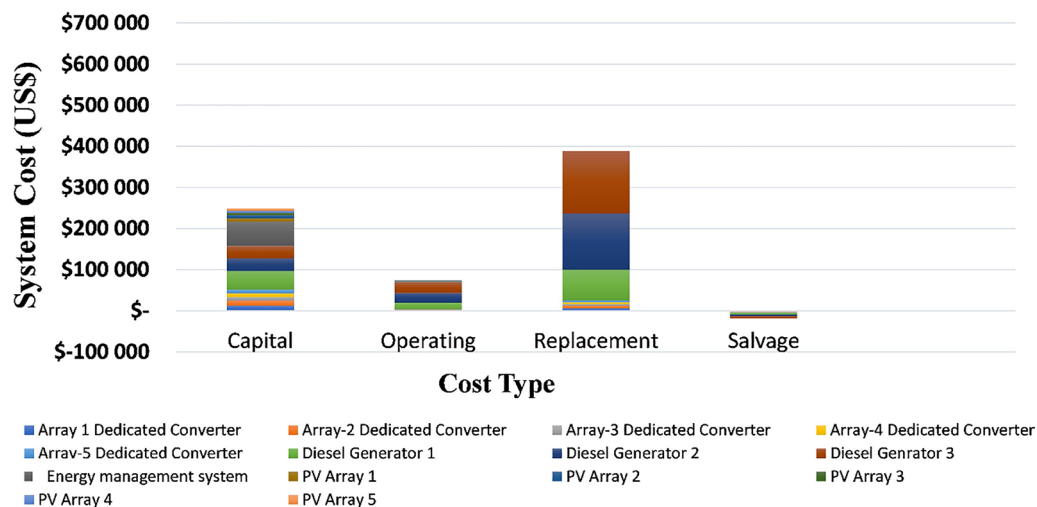


Fig. 8 Total NPC for scenario B

during their operation. The system cost for scenario A is presented in Fig. 8.

PV/diesel/battery configuration (Scenario C)

Scenario C is a PV/diesel/battery hybrid system (see Fig. 3c). The inclusion of storage batteries in the system configuration serves two purposes: storing the excess electricity generated by the system and reducing the operating hours of the diesel generators. The total annual electricity production by the system amounts to 162,940 kWh/year, with about 30% contribution from solar PV arrays and a 70% contribution from diesel generators (see Fig. 9).

The PV/diesel/battery system provides a reliable power supply, meeting 100% of the load demand without any excess electricity generated. The excess electricity produced by the system is stored in batteries, which can

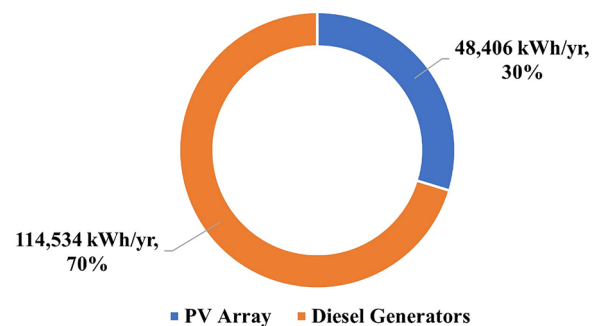


Fig. 9 Total electricity production from scenario C

support the other generating components during high peaks or when the production from the PV array is low. The total renewable fraction is 29.7%, indicating a significant contribution from renewables.

The LCOE obtained from the optimization result is US\$ 0.524/kWh, with a total NPC and operating cost of US\$ 1,177,376 and US\$ 62,889.82, respectively. The annual fuel consumed by the diesel generators is estimated at 36,023 L, and the specific fuel consumption by the three generators (DG1, DG2, and DG3) are found to be 0.270 L/kWh, 0.326 L/kWh, and 0.435 L/kWh, respectively. In addition, the annual CO₂ emission produced by the diesel generators is 94,304 kg/year. The system cost for scenario C is presented in Fig. 10.

Discussion

In the previous section, simulations for the three scenarios have been conducted and the results have been displayed. The results concern techno-economic and environmental parameters in order to search for the most optimal system configuration to meet the electricity demand in Bilgo village. Table 1 shows a comparative analysis of the three scenarios. It can be noted that

the most optimal system with the lowest energy cost and the lowest NPC is scenario C. This scenario consists of a 30 kW_p PV array, a 30 kW inverter, and 56 kW diesel generators combined with a battery storage unit with a capacity of 182 kWh. The LCOE for this scenario is US\$ 0.524/kW, while the NPC is US\$ 1,177,376. Although the initial investment cost in the optimal scenario is high, which is about US\$ 318,818, because of the initial acquisition of components, it has the lowest O & M cost of US\$ 62,890.

Additionally, the optimal system has the lowest replacement cost of components of US\$ 318,023 and the lowest fuel cost of US\$ 510,052 compared to the two other configurations.

From Table 1, one can see that the net present costs for the three scenarios show that scenario B has the highest NPC compared to the other two scenarios due to its high operating cost of US\$ 74,342 and replacement cost of US\$ 364,381. Scenario A has the lowest capital cost of US\$ 165,358 compared to the other two scenarios. It also has the highest fuel consumption cost, totaling US\$ 725,375, due to the extended operating hours of diesel generators.

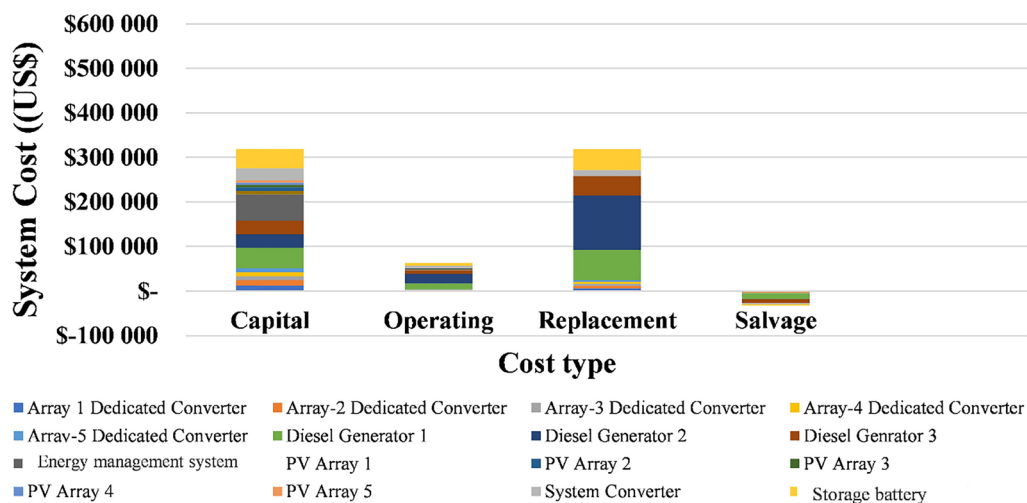


Fig. 10 Total NPC for scenario C

Table 1 Summary of scenarios studied for the power system configuration

System configuration scenario	Electricity production (kWh/year)	Annual fuel consumption (L)	LCOE (US\$ /kWh)	NPC (US\$)	Excess electricity production (kWh/year)	Renewable energy fraction (%)	Annual CO ₂ emission (kg/year)
Diesel generators only (A)	157,154	51,232	0.5816	1,307,126	0	0	134,116
PV/diesel generator (B)	186,877	46,932	0.604	1,357,160	29,732	25.9	122,861.00
PV/diesel/battery (C)	162,940	36,023	0.524	1,177,376	0	29.7	94,304.00

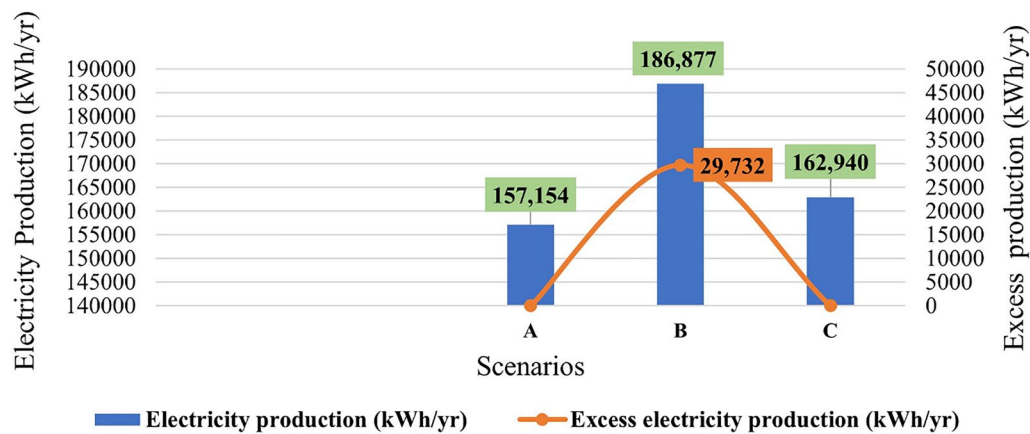


Fig. 11 Comparison of scenarios based on excess energy production and annual electricity production

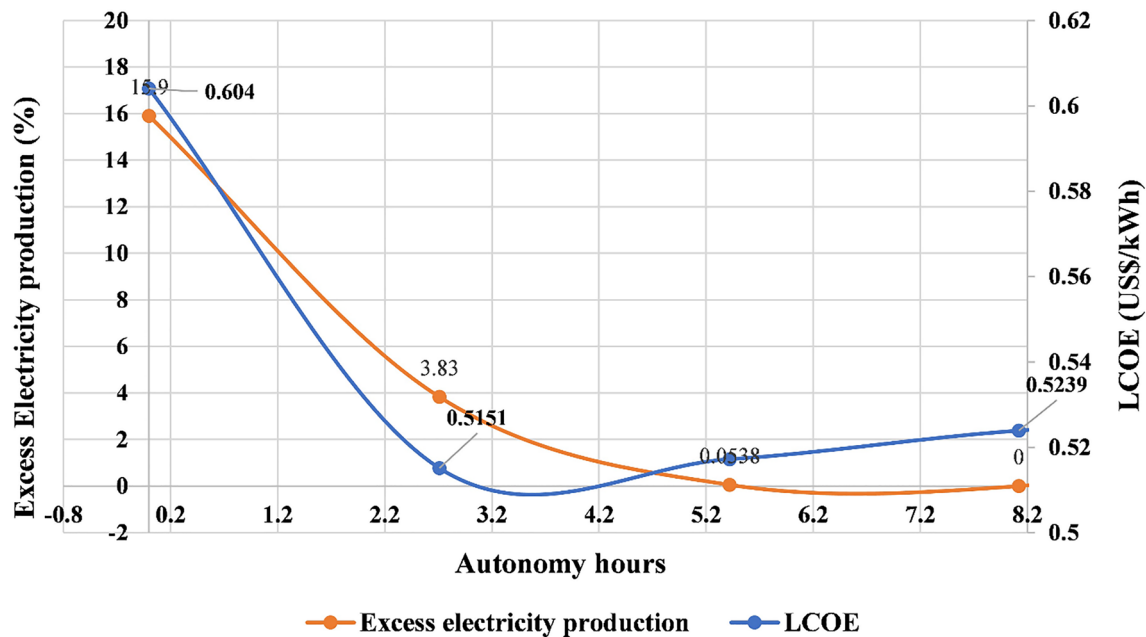


Fig. 12 The effect of battery autonomy hours on LCOE

Electricity production and battery autonomy comparison

The annual electricity production of the optimal system is 162,940 kWh/year. The optimal storage capacity has an autonomy of 8.12 h, which means that it is capable of serving loads during this time. This helps in enhancing system management, as diesel generators can be halted at specific times based on power requirements to meet load demands. In scenario B where there is no storage unit in the system, the excess electricity production is about 16% of the demand and represents a waste of energy. Figure 11 shows the electricity production for the three scenarios.

Figure 12 and Table 2 illustrate the impact of storage autonomy hours on the LCOE. For scenario B (without

Table 2 Impact of storage capacity on the system's LCOE

Battery capacity (kWh)	LCOE (US\$/kWh)	Autonomy (h)	Excess electricity production lost (%)
0	0.604	0	15.9
60	0.515	2.71	3.83
121	0.517	5.42	0.05
182	0.523	8.12	0

storage), it is observed that the LCOE is very high, about US\$ 0.604/kWh. In fact, the LCOE of the scenario with storage is lower than that of the scenario without a

battery (see Table 2). For an autonomy of about 8 h, a battery storage capacity of 182 kWh is required to store all the excess electricity and the management of the system.

Renewable fraction and annual emission produced from the system

The optimal system exhibits the highest renewable energy fraction, approximately 29.6%. It is followed by scenario B with a renewable energy fraction of about 26%. However, scenario A has a renewable energy fraction of zero as it is made entirely of diesel generators. The integration of batteries in the optimal system results in a reduction in the operating hours of the diesel generators by 34.8%.

Moreover, it results in an annual reduction of CO₂ emissions by 29.68% compared to scenario A and by 23.24% compared to scenario B. In scenario B, both annual fuel consumption and CO₂ emissions from the system were lowered by 9.16% and 8.4%, respectively, compared to scenario A, thanks to the contribution of PV. Figure 13 shows a comparison between LCOE and the renewable energy fraction for the three scenarios. Figure 14 shows the comparison based on fuel consumption and CO₂ emissions of each scenario.

Figures 13 and 14 clearly demonstrate the greater technical and environmental efficiency of Scenario C compared with Scenarios A and B.

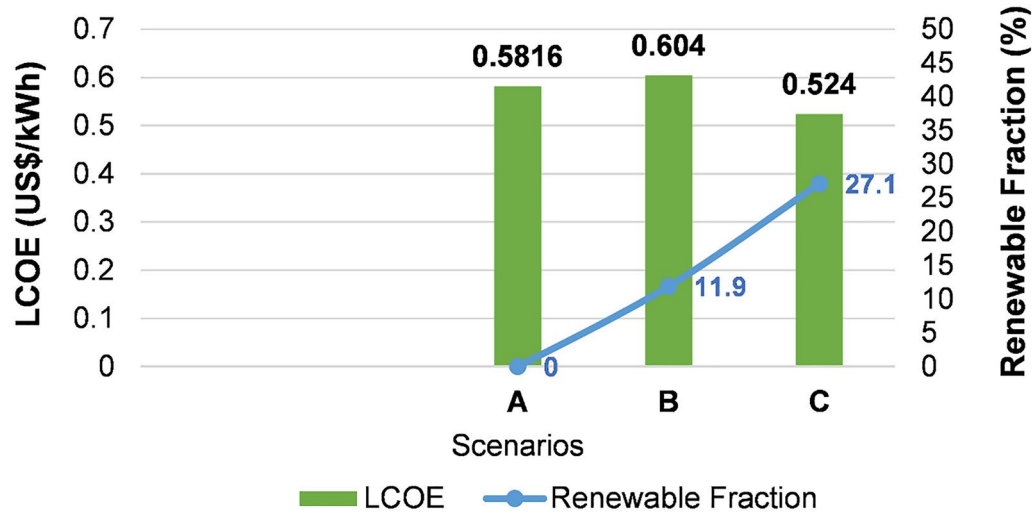


Fig. 13 Scenarios comparison based on LCOE and renewable energy fraction

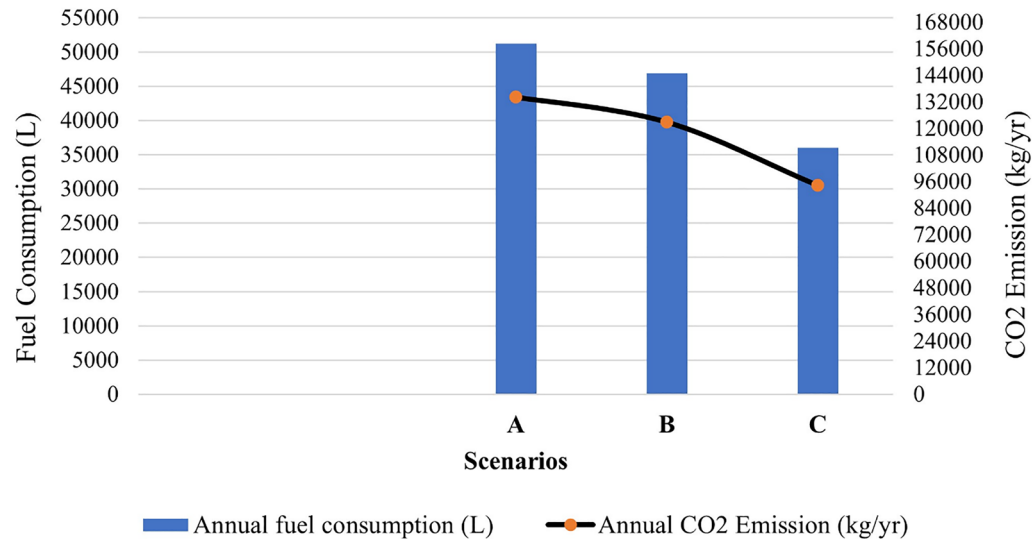


Fig. 14 Scenario-based on fuel consumption and CO₂ emission

Daily energy production for the best scenario

The time resolution of 1 h for system operation used by HOMER allows one to examine the daily variability of the system for a specific scenario. Figure 15 shows the time series plot obtained from the daily production of the diesel generators, PV array, and battery storage for the best scenario (Scenario C) of the present study. It gives a clear idea of how the different generators contribute to meet the daily demand. The average daily electrical output of 132.62 kWh results in a total annual electricity production of 48,406 kWh from the PV array, contributing to a renewable energy penetration of 29.7%. One can also see in Fig. 15 that the operating period of the battery during the day is between 3 and 6 p.m. This operation of batteries contributes to decreasing the LCOE and CO₂ emissions by lowering the fuel consumption of the diesel generators.

Sensitivity analysis

A sensitivity analysis was conducted on the optimal system, considering two distinct fuel price scenarios: US\$ 0.86/L and US\$ 1.28/L. These scenarios were assessed in relation to the current fuel price in Burkina Faso, which stands at US\$ 0.99/L. The results show that when the fuel price decreases from US\$ 0.99/L to US\$ 0.86/L, the LCOE for the optimal system decreases from US\$ 0.524/kWh to US\$ 0.492/kWh which indicates a significant decrease in LCOE by 6.1%. The total NPC falls from US\$ 1,177,376 to US\$ 1,106,786, which represents a reduction

in the NPC by 6%. However, when the price sudden increases from US\$ 0.99/L to US\$ 1.28/L the LCOE of energy rises from US\$ 0.524/kWh to US\$ 0.590/kWh, which represents a 12.6% increase in the LCOE. Similarly, the NPC increases from US\$ 1,177,376 to US\$ 1,326,066, which indicates a rise in the NPC by 12.6%. The results indicate that the LCOE and the NPC depend on the fuel price, that is, if the fuel price increases, the LCOE, and the NPC will increase as well.

Comparison of results with other relevant works in the literature

It can be challenging to compare the results of the present case study with those of other relevant studies in the literature due to the specificity of the context, i.e., the system structures, sizes, and variability of load demands. However, the LCOE can serve as a metric to assess our results against similar efforts in the literature. Table 3 presents a synoptic comparison between the LCOE of the present research and some literature findings.

The tabulated data show that the LCOE of the hybrid system in the case study (Burkina Faso) is the highest among the cases considered. One of the effective parameters that change LCOE are diesel price, interest rate and also fiscal measures in force in the country. It is evident that countries with high fuel prices such as Burkina Faso will have higher LCOE values. Several measures from the government need to be performed to lower the current LCOE in Burkina Faso, such as up-front grants or cash

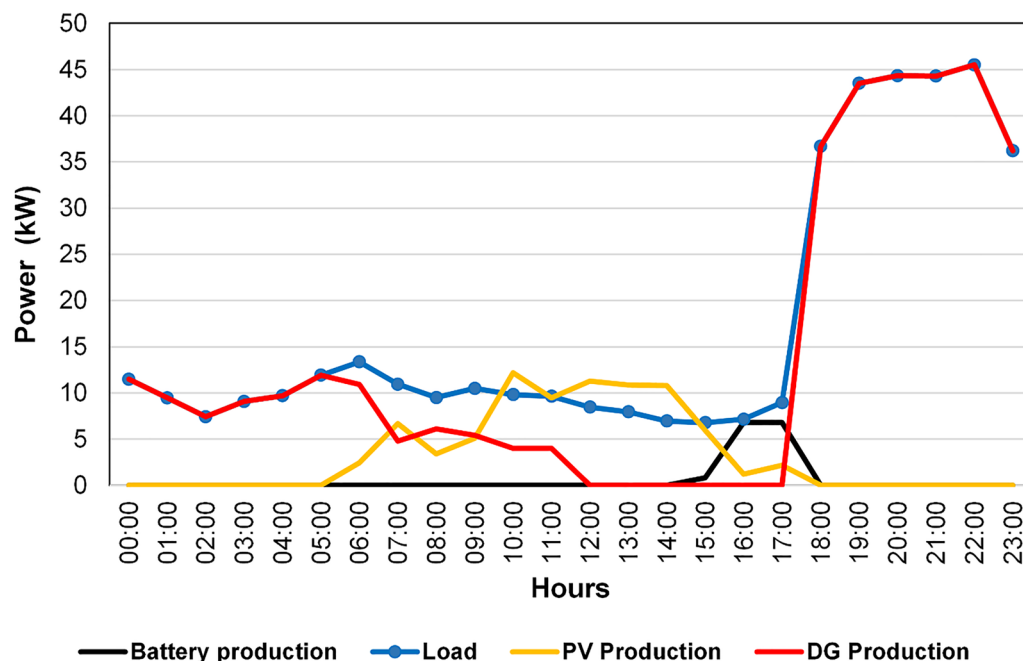


Fig. 15 Daily energy production of the best scenario

Table 3 Comparison of the results of the present study with other relevant studies in the recent literature

References	Geographical context	Optimization method	Hybrid configuration	LCOE (US \$/kWh)	Renewable energy fraction (%)
Oluwarotimi et al. [8]	Benin	HOMER Pro [®]	PV/DG/BAT	0.21	96.7
Oluwarotimi et al. [8]	Benin	HOMER Pro [®]	PV/BAT	0.29	100
Kelly et al. [13]	Chad	HOMER Pro [®]	PV/BAT	0.37	100
Kelly et al. [13]	Chad	HOMER Pro [®]	PV/DG/BAT	0.49	95.5
Kelly et al. [13]	Chad	HOMER Pro [®]	PV/DG/BAT	0.32	82.6
Kelly et al. [13]	Chad	HOMER Pro [®]	PV/BAT	0.36	100
Beza et al. [14]	Ethiopia	HOMER Pro [®]	PV/DG/BAT	0.17	86.4
Beza et al. [14]	Ethiopia	HOMER Pro [®]	PV/BAT	0.26	100
Saleh et al. [15]	Iraq	HOMER Pro [®]	PV/DG/BAT	0.26	44.7
Saleh et al. [15]	Iraq	HOMER Pro [®]	PV/DG/BAT	0.22	18.4
Saleh et al. [15]	Iraq	HOMER Pro [®]	PV/DG/BAT	0.21	35.6
Shezan et al. [16]	Malaysia	HOMER Pro [®]	PV/DG/BAT	0.19	19.9
Shezan et al. [16]	Malaysia	HOMER Pro [®]	PV/DG/BAT	0.21	17.9
Sofmieari et al. [17]	Nigeria	HOMER Pro [®]	PV/DG/BAT	0.49	30
Khamharnphol et al. [18]	Thailand	HOMER Pro [®]	PV/DG/ BAT	0.23	89.4
Mulenga et al. [19]	Zambia	HOMER Pro [®]	DG/PV	0.33	20.5
Mulenga et al. [19]	Zambia	HOMER Pro [®]	PV/DG/BAT	0.18	91.9
Mulenga et al. [19]	Zambia	HOMER Pro [®]	PV/BAT	0.18	100
The present study	Burkina Faso	HOMER Pro [®]	PV/DG	0.61	25.9
The present study	Burkina Faso	HOMER Pro [®]	PV/DG/BAT	0.52	29.7

rebates for installing renewable energy equipments, low interest rate or interest-free loans to organizations or companies that install systems with renewable energy sources. While it is true that the government of Burkina Faso has already adopted a law on free-taxes for solar equipments since 2012 [28], the other measures listed above need to be considered in order to accelerate the rural electrification in the country.

Conclusions

Many studies in the literature focus on PV/diesel hybrid energy systems, yet only a few have addressed the optimization of exiting systems.

The main objective of this paper was to investigate the performance enhancement of the PV/diesel power plant in the Bilgo village in Burkina Faso. The existing power plant lacks battery storage and the optimization approach was able to find an optimal storage capacity that should be included in the system.

This study considered three scenarios: a PV/diesel system, a PV/diesel/battery system, and conventional diesel generators, all applied to the same load profile. The optimization results were examined by evaluating the system configuration using metrics which are LCOE, NPC, CO₂ emissions, excess energy production, and the renewable energy fraction. Based on the obtained results, it can be

noted that the optimal system configuration was found to be the PV/diesel/battery system with an LCOE of US\$ 0.524/kWh. This reflects a significant decrease in energy cost by 13.24% and 9.9% when compared to the scenarios with PV/diesel without battery and the convention diesel generators, respectively.

In terms of financial assessment throughout the system's entire lifetime, the results indicated that the total NPC for the optimal system was reduced by 13.25% and 9.93% in comparison to scenarios PV/diesel without battery and diesel generators, respectively. Regarding the environmental aspect, the annual CO₂ emissions from the optimal system were 29.68% lower than those in the diesel generators scenario. This reduction is due to the lower amount of operating hours of the diesel generators in the optimal scenario. The renewable fraction for the optimal system stands at 27.1%, indicating a high contribution of renewable in the hybrid system. Moreover, the storage unit used in the optimal system configuration is a storage management unit with 8.12 h of autonomy.

In short, the present study clearly shows that, for off-grid rural electrification in Burkina Faso, a hybrid PV/diesel/battery is the most suitable option comparing to PV/diesel and diesel only systems, which are other BAT widely employed technologies for rural electrification in the country. Therefore, it would be appropriate for the

government to take additional measures to incentivize the installation of storage batteries, whose acquisition costs are high, in order to facilitate their integration into hybrid systems and thus accelerate rural electrification in Burkina Faso and more generally in Africa.

Abbreviations

AC	Alternative current
BAT	Battery
DC	Direct current
DG	Diesel generator
HRES	Hybrid renewable energy system
kW	Kilowatt
kWh	Kilowatt-hour
kWp	Kilowatt-peak
LCOE	Levelized cost of electricity
NPC	Net present cost
OM	Operation and maintenance
PV	Photovoltaic

Acknowledgements

The authors are grateful for the support provided by the World Bank Group and the Government of Burkina Faso for their financial support through the Africa Higher Education Centers of Excellence for Development Impact project (IDA 6388-BF / D443-BF). They are also grateful to Mr. Lere Deguenon for the proofreading of the paper.

Author contributions

DY conceived the study, analyzed the data and wrote the first draft of the paper. HTN collected the data and prepared the figures. ASB reviewed the manuscript.

Funding

This work was supported by the African union through the Pan African University Institute of Water and Energy Sciences (Including Climate Change).

Availability of data and materials

The datasets used during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

All the authors read and agree to publish the article.

Competing interests

The authors declare that they have no competing interests.

Author details

¹Laboratoire Energies Renouvelables et Efficacité Energétique (LabEREE), Institut International d'Ingénierie de l'Eau et de l'Environnement (2iE), Rue de la Science, 01 BP 594, Ouagadougou 01, Burkina Faso. ²Pan-African University Institute for Water and Energy Sciences (PAUWES), 119, Pôle Chetouane, Tlemcen 13000, Algeria. ³FasoBiogaz, 13 BP 124, Ouagadougou 13, Burkina Faso.

Received: 16 June 2023 Accepted: 21 July 2024

Published online: 05 August 2024

References

- Mallick TK, Sarmah N, Banerjee SN, Micheli L, Reddy KS, Ghosh PC et al (2013) Design concept and configuration of a hybrid renewable energy system for rural electrification in India through BioCPV project. In: Paper presented at the IVth international conference on advances in energy research, Indian Institute of Technology Bombay, Mumbai, 10–12 December 2013.
- International Energy Agency (2022) SDG7 Database - Data product—IEA. <https://www.iea.org/data-and-statistics/data-product/sdg7-database>. Accessed 13 June 2023.
- International Energy Agency (2018) World energy outlook report. <https://www.iea.org/reports/world-energy-outlook-2018>. Accessed 02 May 2023.
- Borhanazad H, Mekhilef S, Gounder Ganapathy V, Modiri-Delshad M, Mir-taheri A (2014) Optimization of micro-grid system using MOPSO. *Renew Energy* 71:295–306. <https://doi.org/10.1016/j.renene.2014.05.006>
- Chauhan A, Saini RP (2014) A review on integrated renewable energy system based power generation for stand-alone applications: configurations, storage options, sizing methodologies and control. *Renew Sustain Energy Rev* 38:99–120. <https://doi.org/10.1016/j.rser.2014.05.079>
- Shahzad MK, Zahid A, Rashid T, Rehan MA, Ali M, Ahmad M (2017) Techno-economic feasibility analysis of a solar-biomass off grid system for the electrification of remote rural areas in Pakistan using HOMER software. *Renew Energy*. <https://doi.org/10.1016/j.renene.2017.01.033>
- Smith C, Burrows J, Scheier E, Young A, Smith J, Young T et al (2015) Comparative life cycle assessment of a Thai Island's diesel/PV/wind hybrid microgrid. *Renew Energy* 80:85–100. <https://doi.org/10.1016/j.renene.2015.01.003>
- Oluwarotimi ODT, Ramchandra B, Rabani A (2019) Hybrid off-grid renewable power system for sustainable rural electrification in Benin. *Renew Energy* 145:1266–1279. <https://doi.org/10.1016/j.renene.2019.06.032>
- Azoumah Y, Yamegueu D, Ginies P, Coulibaly Y, Girard P (2011) Sustainable electricity generation for rural and peri-urban populations of sub-Saharan Africa: the 'flexy-energy' concept. *Energy Policy* 39(1):131–141. <https://doi.org/10.1016/j.enpol.2010.09.021>
- Boly AS, Yamegueu D, Coulibaly Y (2021) Energy management strategies for PV/diesel hybrid systems in remote areas: effects of 'flexy-energy' strategy. *Int J Renew Energy Res* 11(1):54–65. <https://doi.org/10.20508/ijrer.v11i1.11562.g8171>
- Yamegueu D, Azoumah Y, Py X, Zongo N (2011) Experimental study of electricity generation by Solar PV/diesel hybrid systems without battery storage for off-grid areas. *Renew Energy* 36(6):1780–1787. <https://doi.org/10.1016/j.renene.2010.11.011>
- Tsuanyo D, Azoumah Y, Aussel D, Neveu P (2015) Modeling and optimization of batteryless hybrid PV (photovoltaic)/Diesel systems for off-grid applications. *Energy* 86:152–163. <https://doi.org/10.1016/j.energy.2015.03.128>
- Kelly E, Medjo Nouadje BA, Tonsie Djiela RH, Tiam Kapen P, Ghislain Tchuen G, Tchinda R (2023) Off grid PV/Diesel/Wind/Batteries energy system options for the electrification of isolated regions of Chad. *Heliyon* 9:e13906. <https://doi.org/10.1016/j.heliyon.2023.e13906>
- Beza TM, Wu CH, Kuo CC (2021) Optimal sizing and techno-economic analysis of minigrid hybrid renewable energy system for tourist destination islands of lake Tana. *Ethiopia Appl Sci* 11:7085. <https://doi.org/10.3390/app11157085>
- Saleh AA, Naim Tajuddin MF, Adzman MR, Ramli MAM, Mekhilef S (2019) Energy management and optimization of a PV/diesel/battery hybrid energy system using a combined dispatch strategy. *Sustainability* 11:683. <https://doi.org/10.3390/su11030683>
- Shezan SA, Rawdah, Ali SS, Rahman Z (2021) Design and implementation of an islanded hybrid microgrid system for a large resort center for Penang Island with the proper application of excess energy. *Environ Prog Sustain Energy*. <https://doi.org/10.1002/ep.13584>
- Sofimieari I, Mustafa MWB, Obite F (2019) Modelling and analysis of a PV/wind/diesel hybrid standalone microgrid for rural electrification in Nigeria. *Bull Electr Eng Inf* 8(4):1468–1477. <https://doi.org/10.11591/eei.v8i4.1608>
- Khamharnphol R, Kamdar I, Waewsak J, Chaichan W, Khunpetch S, Chiwamongkhonkarn S et al (2023) Microgrid hybrid Solar/Wind/Diesel and Battery energy storage power generation system: application to Koh Samui, Southern Thailand. *Int J Renew Energy Dev* 12(2):216–226. <https://doi.org/10.14710/ijred.2023.47761>
- Mulenga E, Kabanshi A, Mupeta H, Ndiaye M, Nyirenda E, Mulenga K (2023) Techno-economic analysis of off-grid PV-Diesel power generation system for rural electrification: a case study of Chilubi district in Zambia. *Renew Energy* 203:601–611. <https://doi.org/10.1016/j.renene.2022.12.112>

20. The World Bank (2019) GDP and annual growth rate for Burkina Faso. <https://data.worldbank.org/country/burkina-faso>. Accessed 10 May 2023
21. HOMER (2016) HOMER Pro Version 3.7 user manual. www.homerenergy.com. Accessed 17 Apr 2022
22. Mahmud AM (2016) A post-installation analysis of solar PV-diesel hybrid system for school electrification in Sabah, Malaysia. figshare. <https://hdl.handle.net/2134/24725>. Accessed 17 Apr 2022.
23. SMA (2017) SMA Sunny Island inverter for on-grid and off-grid application. <https://files.sma.de/downloads/SL44M-80H-13-OffGrid-1A-en-10.pdf>. Accessed 17 Apr 2022.
24. SMA Solar Technology (2017) Sunny tripower the three-phase inverter. www.SMA-Solar.com. Accessed 17 Apr 2022/
25. Adaramola MS, Paul SS, Oyewola OM (2013) Assessment of decentralized hybrid PV solar-diesel power system for applications in Northern part of Nigeria. *Energy Sustain Dev*. <https://doi.org/10.1016/j.esd.2013.12.007>
26. Ouedraogo BI, Yamegueu D (2019) Techno-economic assessment of solar photovoltaic integration into national grids : a case study of Burkina Faso. *Energy Sci Eng*. <https://doi.org/10.1002/ese3.364>
27. Global Petrol Price (2019) Diesel prices per litres. https://www.globalpetrolprices.com/diesel_prices/. Accessed 17 June 2019.
28. Assemblée Nationale du Burkina Faso (2012) Loi n°051–2012/AN portant loi de finances pour l'exécution du budget de l'Etat, Gestion 2013. https://www.ceja.ch/images/CEJA/DOCS/Bib/Pays/burkinafaso/loi_n_51_2012_LF_2013_pdf.pdf. Accessed 13 Sept 2023.

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.